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TUTORIAL QUESTIONS

MODULE-II

1. Write a program to remove all vowel characters from a string.

Ans:

```
vowels="AEIOUaeiou"
s=input("Enter the string...")
ns=""
for char in s:
    if char not in vowels:
        ns=ns+char
print("new string after removing vowels=",ns)
```

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2. Write a program to remove characters at odd index positions from a string.

Ans:

```
s=input("Enter the string..:")

i=0

ns=""

while i<len(s):

    if i%2==0:

        ns=ns+s[i]
```

```
    i=i+1
print("New string:",ns)
```

3. Write a python script for palindrome checking without reversing the string.

Ans:

```
s=input("Enter the string...")
if s==s[::-1]:
    print("palindrome..")
else:
    print("not palindrome...")
```

<u>Palindrome checking using loop</u>

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```
s=input("Enter the string..")
beg=0
end=len(s)-1
while beg<end:
   if s[beg]!=s[end]:
        print("Not palindrome")
        break
   beg+=1
   end-=1
```

```
else:
    print("Palindrome")
```

4. Write a program to replace all the spaces in the input string with * or if no spaces found, put $ at the start and end of the string.

Ans:

```
s=input("Enter the string:")
s=s.replace(" ","*")
if "*" not in s:
    s="$"+s+"$"
    print(s)
else:
    print(s)
```

5. Write a program to slice the string into two separate strings; one with all the characters in the odd indices and one with all characters in even indices.

Ans:

```
s=input("enter the string:")
eps=s[0:len(s):2]
print("slice with even position characters:",eps)
ops=s[1:len(s):2]
print("slice with odd position chracters:",ops)
```

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6. Write a program to remove all occurrence of a substring from a string.

Ans:

```
s=input("enter the string..")
ss=input("enter substring to remove..")
ls=len(s) # length of the string
lss=len(ss) # length of the substring
ns="" # new string
i=0
while i<ls:
  css=s[i:lss+i] #css is the substring to be compared extracted from main string
  if css==ss:
    i=i+lss
  else:
    ns=ns+s[i]
    i=i+1
print("new string",ns)
```

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7. Write a Program to converting all lowercase letters into uppercase.

Ans:

```
import string
```

```
s=input('Enter the string...')
ns=""
for c in s:
 if c in string.ascii_lowercase:
    c=chr(ord(c)-32)
 ns=ns+c
print("new string=",ns)
```

8. Write a Program to replace all occurrence of a substring with a new substring.

Ans:

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```
s=input("enter string..")
ss=input("enter substring to remove..")
nss=input("enter the substring to replace....")
ls=len(s)
lss=len(ss)
ns=""
i=0
while i<ls:
  css=s[i:lss+i]
  if css==ss:
```

```
        ns=ns+nss
        i=i+lss
      else:
        ns=ns+s[i]
        i=i+1
    print("new string",ns)
```

9. Write a Program to reverse the first and second half of a string separately.

Ans:

```
    s=input("Enter the string..:")
    l=len(s)
    fs=s[0:l//2]
    ss=s[l//2:]
    fs=fs[::-1]
    ss=ss[::-1]
    s=fs+ss
   print("New string after reversal:::",s)
```

10. Write a Python program to check the validity of a password given by the user

The Password should satisfy the following criteria:

1. Contains at least one letter between a and z
2. Contains at least one number between 0 and 9

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3. Contains at least one letter between A and Z
4. Contains at least one special character from $, #, @

Minimum length of password: 6

Ans:

```
l, u, p, d = 0, 0, 0, 0
s = input(“Create a password )”
if (len(s) >= 6):
    for i in s:
        # counting lowercase alphabets
        if (i.islower()):
            l+=1
        # counting uppercase alphabets
        if (i.isupper()):
            u+=1
        # counting digits
        if (i.isdigit()):
            d+=1
        # counting the mentioned special characters
        if(i=='@'or i=='$' or i=='_'):
```

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```
                p+=1
if (l>=1 and u>=1 and p>=1 and d>=1 and l+p+u+d==len(s)):
    print("Valid Password")
else:
    print("Invalid Password")
```

11.Write Python script for converting decimal number into Binary number.

Ans:

```
decno=int(input("Enter the decimal number...."))
if decno==0:
    print("The binary equivalent is....0000")
else:
    binaryno=""
    while decno!=0:
        b=decno%2
        binaryno=str(b)+binaryno
        decno=decno//2
    print("The binary equivalent is....",binaryno)
```

12.Write Python script for converting Binary number into decimal number.

Ans:

```
bitstring=input("Enter a binary number...")
decno=0
```

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```
    expnt=len(bitstring)-1
    for bit in bitstring:
        decno=decno+int(bit)* 2 ** expnt
        expnt=expnt-1
    print ("The decimal number is=", decno)
```

13.Write a python function to find the area of a circle.

Ans:

```
def circlearea (radius):
    area=3.14*(radius**2)
    return area
#function call
r=int(input("Enter radius.."))
area=circlearea(r)
print("Area of the circle=",area)
```

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14.Write a python program to compute nCr using a factorial function.

Ans:

```
def fact(n):
    f=1
    for i in range(1,n+1):
```

```
        f=f*i
    return f
print("Program to compute nCr...")
n=int(input("Enter n.."))
r=int(input("Enter r..."))
ncr=fact(n)/(fact(n-r)*fact(r))
print("nCr...",ncr)
```

15.Write a menu driven program to implement the following

i)check even or odd

ii)check number is positive negative or zero

iii) generate factors of a number

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Ans:

```
def evenodd(n):
    if n%2==0:
        print("even")
    else:
        print("odd")
def postvnegtv(n):
    if n>0:
        print("+ve")
```

```
    elif n<0:
        print("-ve")
    else:
        print("zero")
def  factors(n):
    print ("factors")
    for i in range(1,n+1):
        if n%i==0:
            print (i,end=' ')
while True:
    print("\n....Menu...\n1.even or odd\n2.postv or negtv \n3.factors\n4..exit\n")
    ch=int(input("Enter your choice---"))
    if ch==4:
        break
    n=int(input("Enter a number.."))
    if ch==1:
        evenodd(n)
    if ch==2:
        postvnegtv(n)
```

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```
    if ch==3:
        factors(n)
```

16. Write a Python program to find the value for sin(x) up to n terms using the series
sin(x)=1-x^3/3!+x^5/5!..... (sin(x) = ((-1)^n/(2n+1)!)x^(2n+1))

Ans:

```
import math
def sinseries(x,n):
    sine = 0
    for i in range(n):
        sign = (-1)**i
        x=x*(math.pi/180)
        sine = sine + ((x**(2.0*i+1))/math.factorial(2*i+1))*sign
    return sine
x=int(input("Enter the value of x in degrees:"))
n=int(input("Enter the number of terms:"))
print(round(sinseries(x,n),2))
```

<u>output</u>:

Enter the value of x in degrees:30

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```
Enter the number of terms:10

0.5
```

17.Write a Python program to print the factorial of a number using recursion.

Ans:

```python
def fact(n):
    if n==0:
        return 1
    else:
        return n*fact(n-1)

n=int(input('Enter n..:'))
x=fact(n)
print("factorial of ..",n ," ..is.. ",x)
```

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18.Write a Python program to print n'th Fibonacci number using recursion.

Ans:

```python
def fib(n):

  if n <= 1:

      return n

  else:

     return fib(n-1) + fib(n-2)
```

```
n=int(input('Enter n…'))

x=fib(n)

print (n,"th Fibonacci number is...",x)
```

19.Program to read list of names and sort the list in alphabetical order.(university question)

Ans:

```
n=int(input("Enter the number of names...."))
names=[]
print("Enter {} names".format(n))
for i in range(n):
    nam=input()
    names.append(nam)
names.sort()
print("names in alphabetical order")
for nam in names:
    print(nam)
```

20.Program to find the sum of all even numbers in a group of n numbers entered by the user. (university question)

Ans:

```
n=int(input("Enter the number of elements..."))
print("Enter the {} elements".format(n))
l=[] # creating an empty list
for i in range(n):
```

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```
    x=int(input())
    l.append(x)
sum=0
for i in range(n):
    if l[i]%2==0:
        sum=sum+l[i]
print("Sum of all even numbers",sum)
```

21.Program to read a string and remove the given words from the string.
Ans:

```
s=input("Enter the string....")
wr=input("Enter the word to remove....")
wrds=s.split(" ")
ns=""
for w in wrds:
    if w!=wr:
        ns=ns+" "+w
print("new string...",ns)
```

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22.Program to read list of numbers and find the median
Ans:

```
#We can find median by sorting the list and then take the middle element. If the
list contains even number of elements take the average of the two middle
elements.

n=int(input("Enter how many numbers...."))
print("Enter {} numbers....".format(n))
```

```
lst=[]
for i in range(n):
   x=int(input())
   lst.append(x)
lst.sort()
print(lst)
mid=n//2
if n%2==1:
   print("Median",lst[mid])
else:
   print("Median",(lst[mid]+lst[mid-1])/2)
```

23.Finding the mode of list of numbers(A number that appears most often is the mode.)

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Ans:

```
l=[]
n=int(input("Enter n.."))
print("Enter the numbers..")
for i in range(n):
    x=int(input())
    l.append(x)
c=[]
e=[]
for x in l:
    if x not in e:
        c.append(l.count(x))
```

```
        e.append(x)
    mc=max(c)
    ne=len(c)
    i=0
    print("mode..")
    while i<ne:
        if c[i]==mc:
            print(e[i])
        i+=1
```

24.Program to remove all duplicate elements from a list
Ans:

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```
lst=[]
n=int(input("enter how many numbers.."))
print("Enter elements...")
for i in range(n):
    x=int(input())
    lst.append(x)

nlst=[]
for x in lst:
    if x not in nlst:
        nlst.append(x)
print("new list after removing duplicates")
print(nlst)
```

25.Consider a list consisting of integers, floating point numbers and strings. Write a program to separate them into different lists depending on the data(university question)

Ans:

```
il=[]
fl=[]
sl=[]
l=[12,23.5,'klf',34,4343,34.566,3+3j,'ddldl',35]
for i in l:
  if type(i)==int:
    il.append(i)
  if type(i)==str:
    sl.append(i)
  if type(i)==float:
    fl.append(i)
print("Integer list")
print(il)
print("Float list")
print(fl)
print("String List")
print(sl)
```

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26.Write a Python program to read list of positive integers and separate the prime and composite numbers (university question).

Ans:

```
def prime(n):
        flag=1
```

```
        for i in range(2,n//2+1):
                if n%i==0:
                        flag=0
                        break
        return flag
pl=[]
cl=[]
l=[]
n=int(input("Enter n.."))
print("Enter the numbers..")
for i in range(n):
    x=int(input())
    l.append(x)
for i in l:
  if prime(i):
        pl.append(i)
  else:
        cl.append(i)
print("prime list")
print(pl)
print("composite list")
print(cl)
```

27. Write a Python program to read a list of numbers and sort the list in a non-decreasing order without using any built in functions. Separate function should be written to sort the list wherein the name of the list is passed as the parameter.

Ans:

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```
def sortlist(lst,n):
    for i in range(n-1):
        for j in range(i+1,n):
            if lst[i]>lst[j]:
                lst[i],lst[j]=lst[j],lst[i]
n=int(input("Enter how many numbers...."))
print("Enter {} numbers....".format(n))
lst=[]
for i in range(n):
    x=int(input())
    lst.append(x)
sortlist(lst,n)
print("sorted List is")
print(lst)
```

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28.Write a program to do basic set operations.

Ans:

```
na=int(input('Enter number of elements of the set A ..'))
A=set()
print("Enter the elements of set A...")
for i in range(na):
    x=int(input())
    A.add(x)
nb=int(input('Enter number of elements of the set B ..'))
B=set()
print("Enter the elements of set B...")
for i in range(nb):
```

```
    x=int(input())
    B.add(x)
print("set operations..")
print("union")
print(A|B)
print("Intersection")
print(A&B)
print("Difference A-B and B-A")
print(A-B)
print(B-A)
print("symmetric Difference")
print(A^B)
```

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29.Program to Remove duplicate elements from a list .

Ans:

```
lst=[]
n=int(input("enter how many numbers.."))
print("Enter elements...")
for i in range(n):
    x=int(input())
    lst.append(x)
nlst=list(set(lst))
print("new list after removing duplicates")
print(nlst)
```

30.Program to completely remove duplicate elements without keeping any copy.

Ans:

```
n=int(input('Enter number of elements of the list ..'))
lst=[]
print("Enter the elements...")
for i in range(n):
    x=int(input())
    lst.append(x)
nlst=list(set(lst)) # new list with only one copy
nlwd=[] # new list without duplicates
for i in nlst:
    if lst.count(i)==1: #non duplicate element
        nlwd.append(i)
print("New list after removing duplicates completely...")
print(nlwd)
```

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31. Program to count the number of occurrence(frequency) of each letters in a given string(histogram)

Ans:

```
S=input("Enter the string…..")
d=dict()
for c in S:
    d[c]=d.get(c,0)+1
print("letter count")
print(d)
```

32. Program to display the frequency of each word in a given string.(university qstn)

Ans:

```
S=input("Enter the string…..")
S=S.split(" ") #splitting it into words
d=dict()
for w in S:
    d[w]=d.get(w,0)+1
print("word count")
print(d)
```

33. Write a Python program to create a dictionary of roll numbers and names of five students. Display the names in the dictionary in alphabetical order.(university question)

Ans:

```
d={}
for i in range(5):
    rn=int(input("Enter roll number.."))
    name=input("Enter name …")
    d[rn]=name

l=list(d.items())
l.sort(key=lambda v:v[1])
print("name and roll number in sorted order of name")
for i in l:
    print(i[1],":",i[0])
```

34. Program to read name and phn numbers of ‘n’ customers and print the list in sorted order of names.

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Ans:

```
n=int(input("Enter number of customers.."))
d={}
for i in range(n):
    nm=input("Enter name..")
    phn=int(input("Enter phn number…"))
    d[nm]=phn

l=list(d.items()) # creating a list
l.sort() # sorting the list in the order of name..
          #l.sort(key=lambda v:v[1]) will sort in the order of phone number
print("name and phn number in sorted order")
for i in l:
    print(i[0],":",i[1])
```

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35. Write a program that uses a dictionary to convert hexadecimal number into binary.

Ans:

```
hextobin={'0':'0000','1':'0001','2':'0010','3':'0011','4':'0100','5':'0101','6':'0110','7':'
0111','8':'1000','9':'1001','A':'1010','B':'1011','C':'1100','D':'1101','E':'1110','F':'11
11'}
n=input('Enter the hexadecimal number….')
bn=''
n=n.upper()
for d in n:
    h=hextobin.get(d)
```

```
        if h==None:
            print('Invalid Number')
            break
        bn=bn+hextobin[d]
    else:
        print('Binary equivalent is..',bn)
```

36.Finding the mode of list of numbers

Ans:

```
    n=int(input("Enter..how many numebrs.."))
    print("Enter {} numbers".format(n))
    numbers=[]
    for i in range(n):
        x=int(input())
        numbers.append(x)
    ncount={}
    for x in numbers:
        ncount[x]=ncount.get(x,0)+1
    maxcount=max(ncount.values())
    print('Mode..')
    for k in ncount:
        if ncount[k]==maxcount:
            print(k)
```

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37. Write a Python code to create a function called list_of_frequency that takes a string and prints the letters in non-increasing order of the frequency of their occurrences. Use dictionaries.

Ans:

```
def list_of_frequency(s):
    d=dict()
    for c in s:
        d[c]=d.get(c,0)+1
    print("letter count in the decresing order")
    l=list(d.items())
    l.sort(key=lambda x:x[1],reverse=True)
    print(l)

s=input("Enter the string….")
list_of_frequency(s)
```

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